# CLOUD SECURITY INDIVIDUAL REPORT

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## Introduction

In today's digital era, cloud security plays a vital role in safeguarding sensitive data and ensuring the integrity and availability of information in cloud environments. As part of my practical experience in this internship program, I participated in a cloud security sprint focused on Microsoft Azure. The project was conducted under the pseudo-organizational name **Testech**, which had been previously adopted in earlier sprints of this internship program, where I was assigned the role of Head of Finance within the team. The sprint was led by a designated team leader who created the Azure account for our pseudo organization and assigned access credentials to all members, including myself.

As the Head of Finance, my responsibilities centered on financial data migration and cloud budget oversight, both critical aspects in maintaining financial integrity and ensuring controlled expenditure in cloud operations. My access privileges allowed me full control over billing systems and financial data, with read-only access to usage reports and project logs, reflecting a realistic implementation of role-based access control (RBAC) in cloud environments.

One of the core components of the sprint for me was familiarizing myself with Microsoft Sentinel, a powerful cloud-native Security Information and Event Management (SIEM) and Security Orchestration Automated Response (SOAR) solution. I gained insight into its relevance in cloud threat detection, proactive incident response, and security automation.

As a team, we collaboratively created a resource group on Azure and successfully deployed a virtual machine (VM). These exercises provided hands-on experience with infrastructure provisioning in the cloud. Furthermore, I was introduced to the concept of Data Loss Prevention (DLP) and its significance in protecting sensitive organizational data from leaks or unauthorized access. As a team, we created and configured DLP permissions through Microsoft Purview (A comprehensive suite of data governance, security, and compliance solutions that helps organizations manage and protect their data across various environments) reinforcing our understanding of compliance and data governance within a cloud ecosystem.

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## Lessons Learnt

### 1. Role-Based Access Control (RBAC) and Identity Management

A significant lesson I learnt was how Azure's RBAC can effectively define and limit user actions based on organizational roles. As Head of Finance, I was given broad access to financial records and billing systems, yet access to usage reports and logs was read-only. This structure ensures that sensitive data is available only to authorized personnel, minimizing risks of data tampering and maintaining accountability.

**2. Microsoft Sentinel and Cloud Threat Detection**

My introduction to Microsoft Sentinel revealed its crucial role in modern cloud security. Sentinel's ability to collect data across various Azure resources, analyze patterns, and provide automated responses to security threats illustrated the importance of proactive defence strategies. I learned how machine learning models and automation rules are used to detect anomalies and streamline incident response processes.

**3. Infrastructure Provisioning and Resource Management**

Working with the team to deploy a virtual machine within a resource group highlighted practical skills in resource provisioning, network configuration, and access control policies. Understanding how to set up and manage a virtual machine environment has given me foundational insight into cloud infrastructure management, which is critical in both security and operational efficiency.

**4. Data Loss Prevention and Compliance via Microsoft Purview**

The concept of Data Loss Prevention (DLP) was particularly enlightening. I learned that DLP strategies are implemented to prevent sensitive information (such as credit card numbers or financial data) from being exposed accidentally or maliciously. Through Microsoft Purview, we applied permissions and DLP rules to ensure that data access and movement within the organization comply with internal policies and external regulations.

**5. Hands-on Access and Basic VM Networking Commands**

In addition to deploying the virtual machine, I learned how to remotely access the VM using the Azure portal and command-line interface. This allowed me to interact with the system and perform basic but essential network diagnostics. For example, we used the "ifconfig" command to check the IP address and DNS configuration of the virtual machine. This practical exposure helped solidify my understanding of how virtual machines operate within cloud networks, and how to verify or troubleshoot basic connectivity issues.

**6. Team Collaboration and Task Execution**

This sprint reinforced the importance of team collaboration in cybersecurity operations. Each member had a defined role and responsibilities, and working together was essential in deploying resources and configuring security settings. The shared tasks, such as creating DLP rules and deploying virtual machines, allowed us to combine knowledge and leverage diverse perspectives to solve complex problems effectively.

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## Conclusion

The cloud security sprint at Testech provided a comprehensive overview of the essential principles and practices involved in securing cloud environments. My role as Head of

Finance gave me hands-on exposure to financial data management, cloud budgeting, and security best practices for protecting sensitive organizational data.

The practical introduction to Microsoft Azure, including working with Microsoft Sentinel and Microsoft Purview, has significantly improved my understanding of cloud-native security tools and governance policies. I now appreciate the critical role of Data Loss Prevention, SIEM tools, and RBAC in maintaining a secure and compliant cloud infrastructure.

Moreover, the experience underscored the value of collaborative teamwork in executing security strategies and deploying cloud resources efficiently. These lessons will undoubtedly inform my future decisions and actions in professional cybersecurity settings, especially in environments where cloud computing continues to evolve as the backbone of modern IT systems.